Introduction to Blockchain

Basic Guide for Beginners

BS CS – F24 – Sec A

Department: Information Engineering Technology

Application of Information and Communication Technology

Authors

Sumama Rahim

Bilal Afridi

M. Faik

Table of Contents

Introduction to Blockchain 3
Why Blockchain? 3
Types of Blockchain 3
i. Private Blockchain 3
ii. Public Blockchain 4
iii. Hybrid Blockchain 4

Group Members:

Bilal Afridi

Muhammad Faik

Sumama Rahim

Introduction to Blockchain

The simple definition: It is a software, which stores record of transections permanently, it is like a ledger. It is open and distributed to record transactions between 2 parties efficiently, verifiable and permanently. We can say that blockchain built trust between 2 parties by making a copy of each transaction and sending that to all parties to make sure everyone know each other and every transaction is known and verifiable, hence it is decentralized system and no middle man or system involved to control it. Having a copy of each transaction with everyone to make sure the security and privacy as well.

…

Technical definition: It is an open and distributed ledger to record transaction which is encrypted and can be read only by computers not human. It is a combination of Cryptography and Human Logic. (**Crypto** means Secret and **Graphy** means Writing/Text)

…

Why Blockchain?

It built trust and it have transparency of data like everyone owns their data and with their own control. To track real time transactions we need blockchain to implement. To make copies of data or duplicating or altering we don't have any option and we also we can't change or update or delete or reverse existing record. As we talk earlier that blockchain is decentralized so there is no third party is involved like agencies, government, or other concerned party/ies. Every record must be verified by each node on the network to build trust and efficiency and every node can verify any record anytime. Every company or organization is looking to reduce the cost of production, so blockchain is definitely can help in order to reduction of cost. As we know the implementation cost is very high but letter it will save more amount because everything will be done automatically.

Types of Blockchain

There are mainly three types of blockchain in current technology era. To join any blockchain network you need a special software to operate your system/computer etc as a blockchain node.

"Node is a system or computer which is connected to blockchain network. And it has three types itself (Half/Lite Node, Full Node, Miner Node)."

i. Private Blockchain

This network can join only by the persons who is authorized and belonging to that organization who are building their blockchain.

ii. Public Blockchain

This network can join by any person who want to be a part of it. It don’t need any authorization to become part of this network.

iii. Hybrid Blockchain

Hybrid blockchain network is a combination of Public and Private Blockchain Network. In this network limited access is with public but mainly and full access is with only the organization.